PRACTICAL 6: LAN CONSTRUCTION AND SWITCH CONFIGURATION

Objective

Build a LAN using a Cisco Catalyst switch.

Equipment

- 10 PCs
- Cisco Catalyst 3560 Switch
- Patch Panel
- Fly Leads

Activities

1. Connect all 10 PCs.

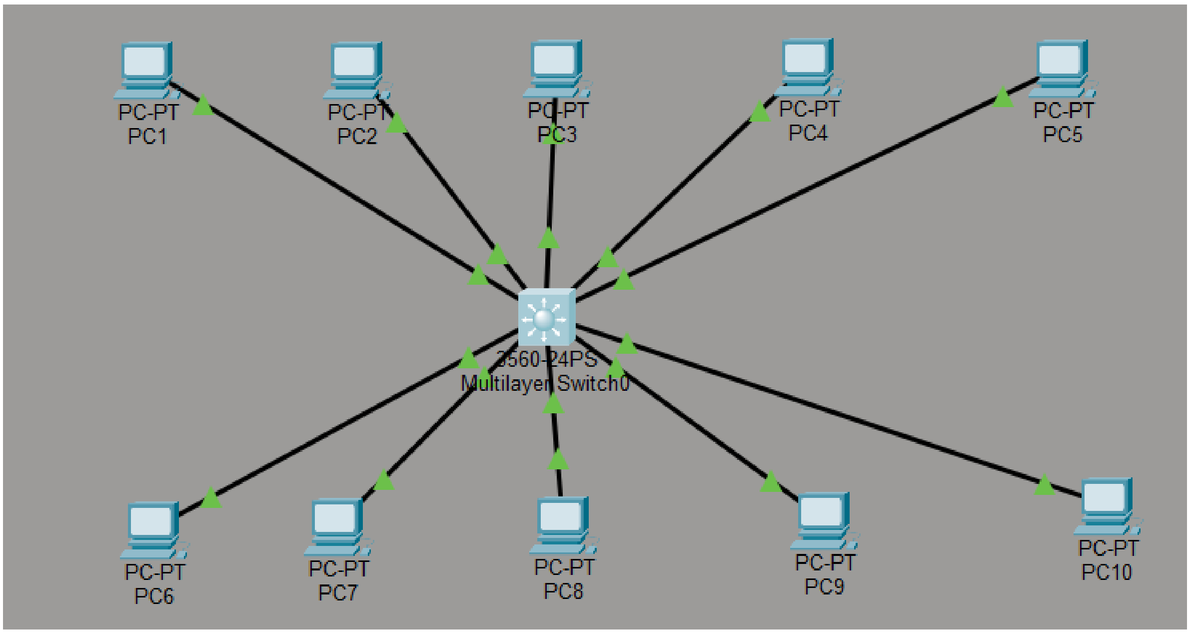


Figure 1: 10 pc's connected

2. Configure static IP addresses.

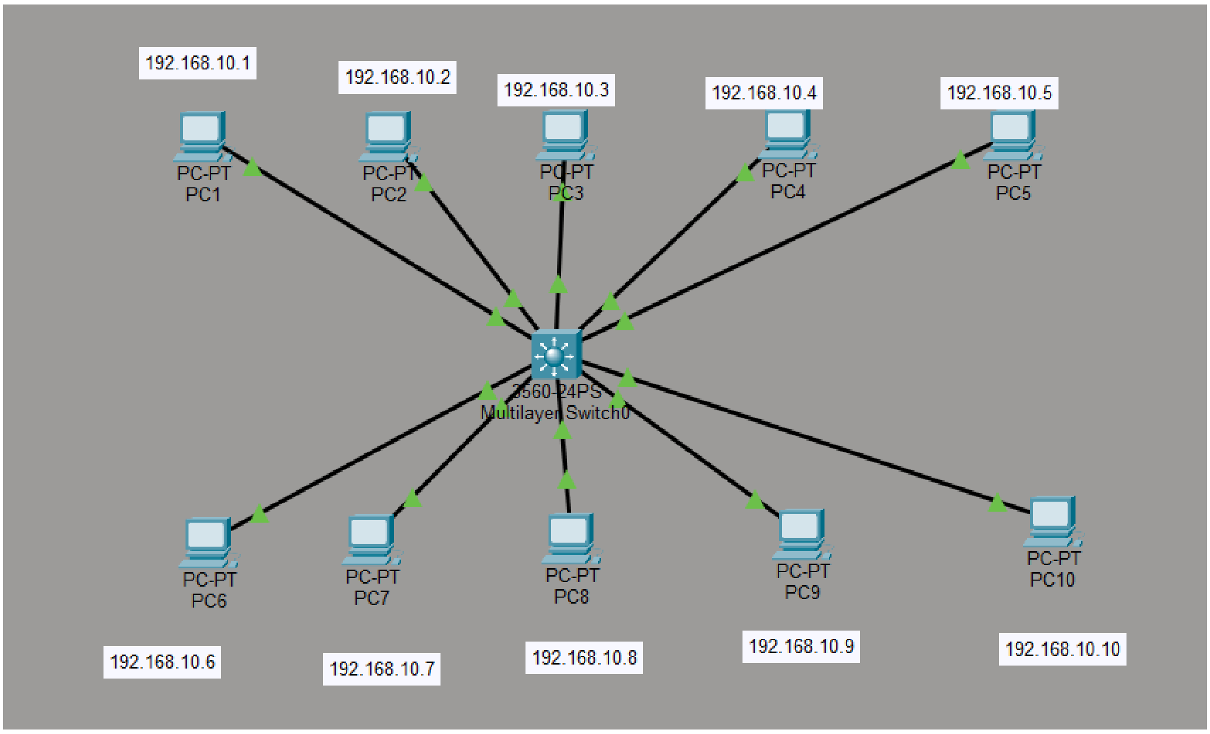


Figure 2: Static IP adressess configured

3. Configure switch management interface.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.10.254 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch(config-if)#ip default-gateway 192.168.10.1
Switch(config)#
```

Figure 3: Switch config

4. Verify connectivity.

```
interface Vlan1
 ip address 192.168.10.254 255.255.255.0
!
ip default-gateway 192.168.10.1
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
!
end
```

Figure 4: show running-config on switch

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Figure 5: Show vlan brief on switch

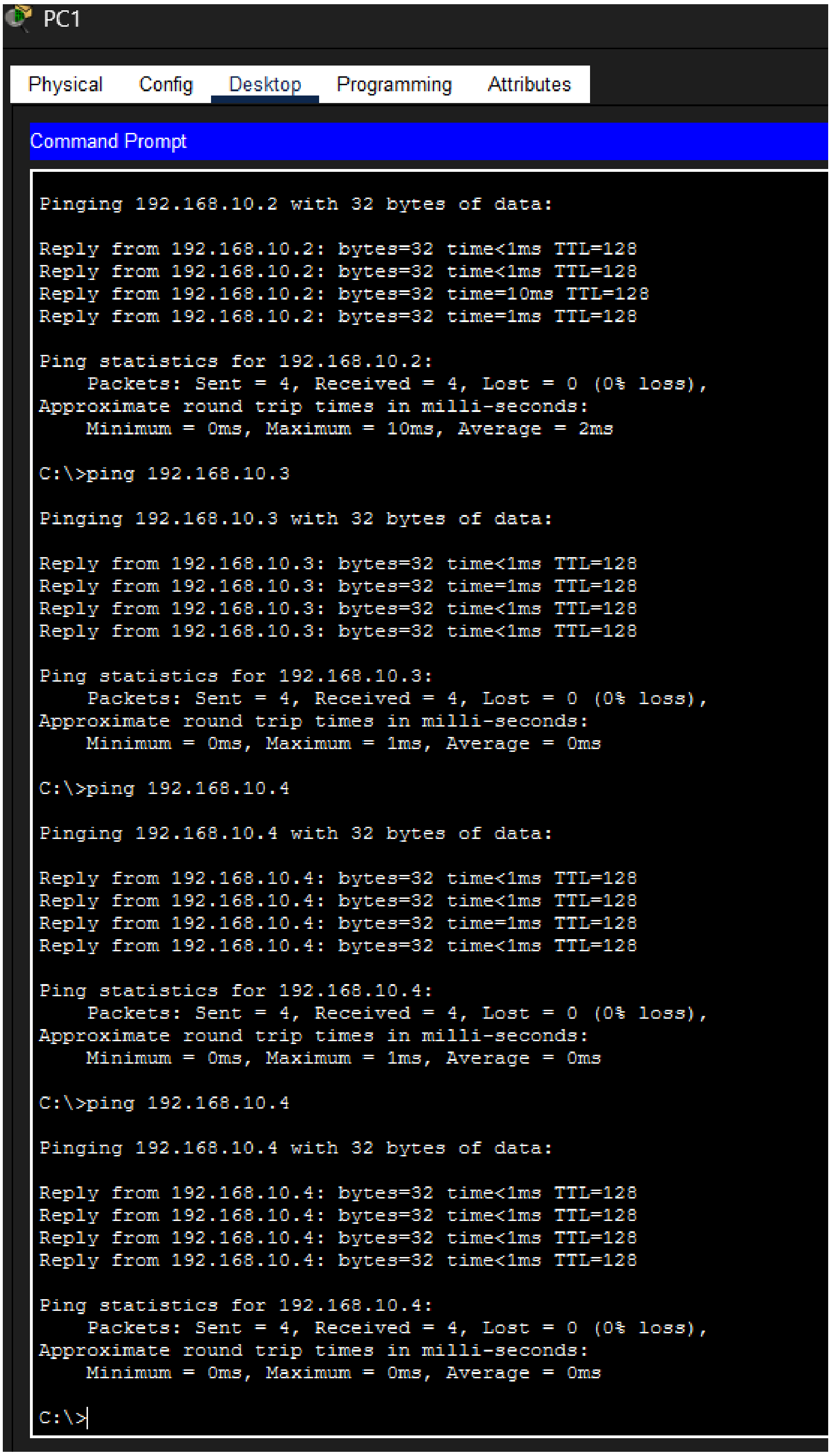


Figure 6: ping from PC1 to PC2-4

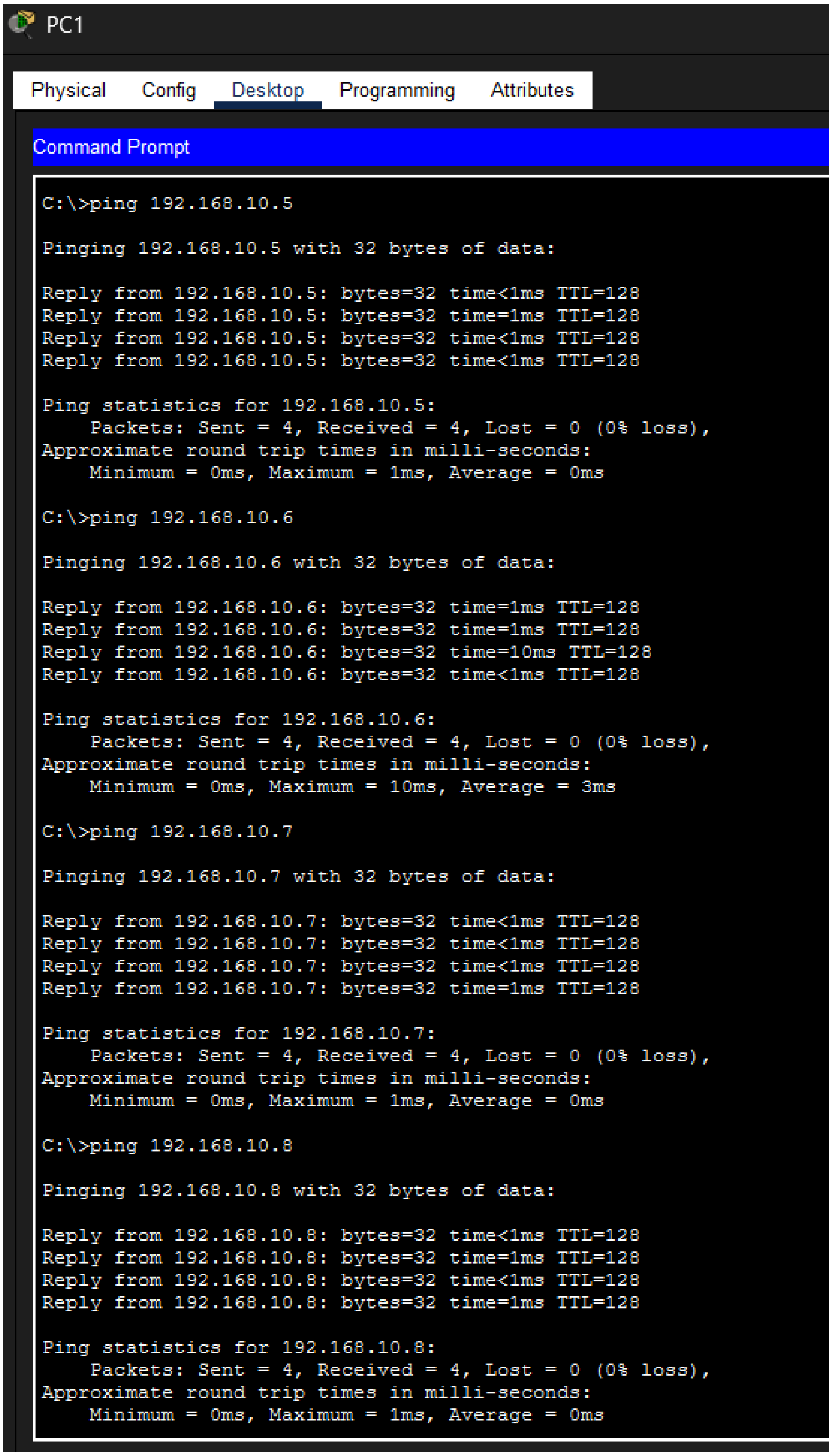
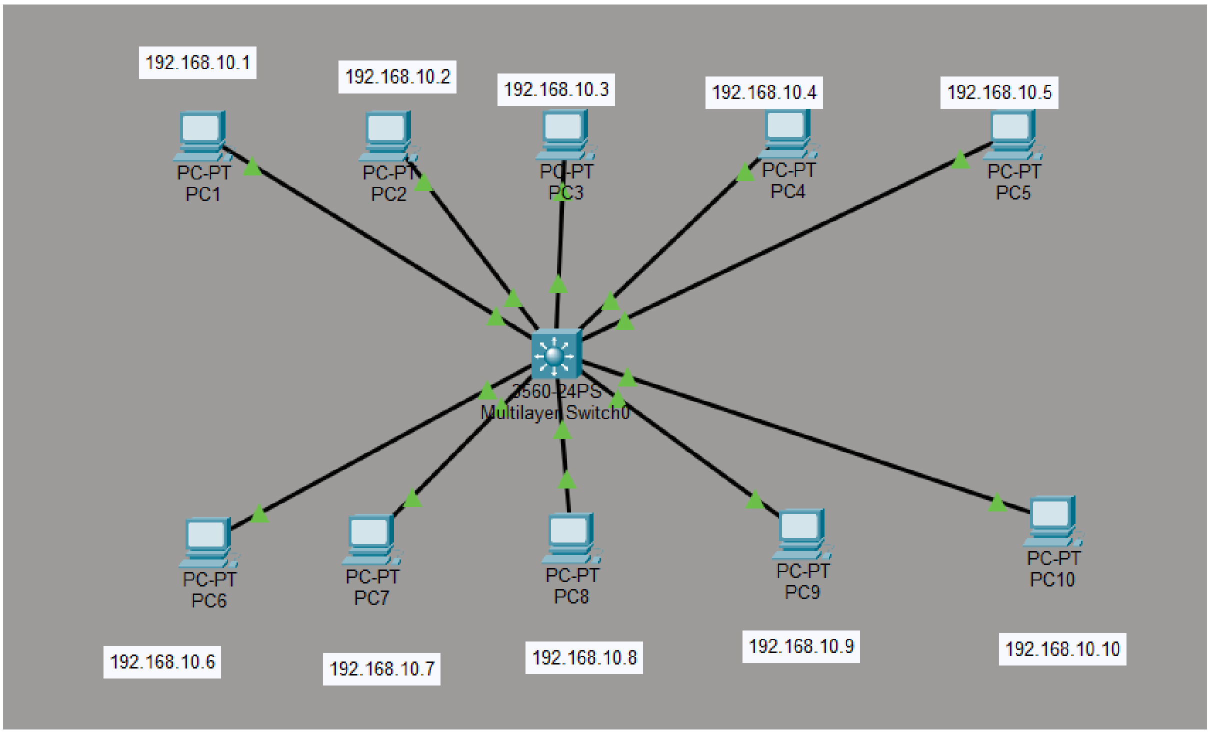


Figure 7: ping from PC1 to PC5-8

```
C:\>ping 192.168.10.9

Pinging 192.168.10.9 with 32 bytes of data:

Reply from 192.168.10.9: bytes=32 time=1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 8: ping from PC1 to PC9-10

5. Create a network diagram.



IP Addressing Scheme **Device**	**IP Address**
PC1	192.168.10.1

PC2	192.168.10.2
PC3	192.168.10.3
PC4	192.168.10.4
PC5	192.168.10.5
PC6	192.168.10.6
PC7	192.168.10.7
PC8	192.168.10.8
PC9	192.168.10.9
PC10	192.168.10.10
Switch VLAN 1	192.168.10.254